

Assignment 4 Report

Pets Expression Classification

1 Experimental Setup

Dataset Classes: Angry, Other, Sad, Happy

Training Images	1000
Validation Images	36
Testing Images	38

Table 1: Dataset Split

Training Settings

- Image Size: 224×224
- Epochs: 30
- Batch Size: 32
- Optimizer: Adam
- Weight Decay: 5×10^{-4}
- Loss Function: CrossEntropy
- Scheduler: CosineAnnealingLR

Data Augmentation

- Resize to 224×224
- Random Horizontal Flip ($p = 0.5$)
- Random Rotation (15°)
- Color Jitter
- Random Affine Scaling (0.9–1.1)
- ImageNet Normalization

2 Models Trained From Scratch

2.1 VGG19

Train Acc.	Val Acc.	Test Acc.	Weighted F1
30.70%	41.67%	39.47%	0.37

Table 2: VGG19 Performance

$$\begin{bmatrix} 4 & 0 & 0 & 6 \\ 2 & 1 & 0 & 3 \\ 2 & 0 & 2 & 7 \\ 1 & 0 & 2 & 8 \end{bmatrix}$$

Table 3: VGG19 Confusion Matrix

Moderate performance with a tendency to classify several samples as Happy.

2.2 ResNet18

Train Acc.	Val Acc.	Test Acc.	Weighted F1
39.00%	44.44%	36.84%	0.33

Table 4: ResNet18 Performance

$$\begin{bmatrix} 2 & 2 & 2 & 4 \\ 1 & 0 & 2 & 3 \\ 0 & 0 & 5 & 6 \\ 1 & 0 & 3 & 7 \end{bmatrix}$$

Table 5: ResNet18 Confusion Matrix

ResNet18 struggled particularly with the Other class.

2.3 DenseNet121

Train Acc.	Val Acc.	Test Acc.	Weighted F1
36.10%	44.44%	42.11%	0.40

Table 6: DenseNet121 Performance

$$\begin{bmatrix} 2 & 1 & 1 & 6 \\ 0 & 2 & 1 & 3 \\ 1 & 2 & 4 & 4 \\ 1 & 2 & 0 & 8 \end{bmatrix}$$

Table 7: DenseNet121 Confusion Matrix

DenseNet121 achieved the strongest results among scratch-trained models.

2.4 MobileNet

Train Acc.	Val Acc.	Test Acc.	Weighted F1
30.20%	47.22%	26.32%	0.22

Table 8: MobileNet Performance

$$\begin{bmatrix} 1 & 0 & 5 & 4 \\ 0 & 0 & 4 & 2 \\ 1 & 0 & 6 & 4 \\ 1 & 0 & 7 & 3 \end{bmatrix}$$

Table 9: MobileNet Confusion Matrix

MobileNet produced the weakest testing performance.

2.5 InceptionV3

Train Acc.	Val Acc.	Test Acc.	Weighted F1
37.10%	50.00%	31.58%	0.28

Table 10: InceptionV3 Performance

$$\begin{bmatrix} 2 & 0 & 5 & 3 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 6 & 5 \\ 1 & 0 & 6 & 4 \end{bmatrix}$$

Table 11: InceptionV3 Confusion Matrix

Higher validation accuracy did not generalize to the test set.

3 Discussion

Training deep CNN architectures from scratch using only 1000 training images proved challenging. Although augmentation and learning-rate scheduling were used, the models generally exhibited limited generalization and noticeable class confusion.

Large architectures such as VGG, ResNet, and DenseNet require significantly larger datasets to learn strong feature representations from random initialization. The obtained results suggest that the available dataset size was insufficient for fully exploiting the capacity of these networks.

4 Transfer Learning

DenseNet121 was selected for transfer learning due to its superior scratch-trained performance.

The transfer-learning pipeline used:

- ImageNet-pretrained DenseNet121
- Backbone freezing
- Fine-tuning of DenseBlock4
- Dropout(0.5)+Linear classifier
- Class-weighted CrossEntropy
- Adam optimizer
- ReduceLROnPlateau scheduler

4.1 DenseNet121 Transfer Learning

Train Acc.	Val Acc.	Test Acc.	Weighted F1
99.40%	44.44%	55.26%	0.52

Table 12: DenseNet121 Transfer Learning Performance

$$\begin{bmatrix} 7 & 1 & 1 & 1 \\ 2 & 0 & 4 & 0 \\ 2 & 0 & 9 & 0 \\ 3 & 1 & 2 & 5 \end{bmatrix}$$

Table 13: Transfer Learning Confusion Matrix

Transfer learning produced the best overall performance, improving test accuracy from 42.11% to 55.26% and weighted F1 from 0.40 to 0.52.

5 Conclusion

DenseNet121 achieved the best results among scratch-trained models. However, the limited dataset size constrained overall performance. Transfer learning using ImageNet-pretrained DenseNet121 substantially improved generalization and achieved the strongest classification performance.